Example 6.16 Find the area of the right angled triangle with hypotenuse 5*cm* and one of the acute angle is $48°30'$

Solution

From the figure,

$$\sin\theta = \frac{AB}{AC}$$

$$\sin 48°30' = \frac{AB}{5}$$

$$0.7490 = \frac{AB}{5}$$

$$5 \times 0.7490 = AB$$

$$AB = 3.7450\ cm$$

$$\cos\theta = \frac{BC}{AC}$$

$$\cos 48°30' = \frac{BC}{5}$$

$$0.6626 = \frac{BC}{5}$$

$$0.6626 \times 5 = BC$$

$$BC = 3.313\ cm$$

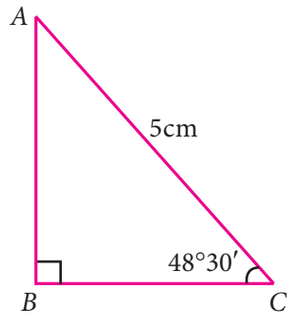


Fig. 6.14

$$\text{Area of right triangle} = \frac{1}{2}bh$$

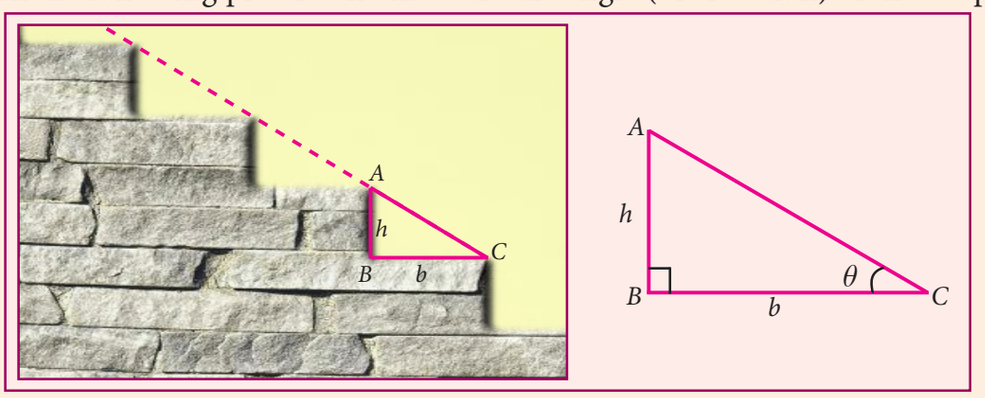
$$= \frac{1}{2} \times BC \times AB$$

$$= \frac{1}{2} \times 3.3130 \times 3.7450$$

$$= 1.6565 \times 3.7450 \quad = 6.2035925\ cm^2$$

Activity

Observe the steps in your home. Measure the breadth and the height of one step. Enter it in the following picture and measure the angle (of elevation) of that step.



(i) Compare the angles (of elevation) of different steps of same height and same breadth and discuss your observation.

(ii) Sometimes few steps may not be of same height. Compare the angles (of elevation) of different steps of those different heights and same breadth and dicuss your observation.

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